

CODE EXPLAINED · BEGINNER FRIENDLY

Grade Calculator

— How It Works

A plain-English walkthrough of a Python script that works out your current grade and tells you exactly what score you need on your final exam to hit your target.

What This Program Does

IN ONE SENTENCE

The program looks at your assignment, quiz, and midterm scores, works out your grade so far, and then calculates the final exam score you need to reach a grade you choose.

WHY IT'S USEFUL

- No manual math or guesswork
- Automatically drops your worst quiz
- Tells you if your target is even possible

The Starting Information

ITEM	VALUE	WHAT IT MEANS
Assignments	88, 94, 91, 85	Four assignment scores (out of 100)
Quizzes	9, 10, 8, 7, 9	Five quiz scores (out of 10 each)
Midterm	82	One midterm score (out of 100)

How Much Each Part Counts

CATEGORY	WEIGHT	SHARE OF FINAL GRADE
Assignments	0.25	25%
Quizzes	0.20	20%
Midterm	0.25	25%
Final Exam	0.30	30%

Note: These four weights add up to 1.00 (100%) — that's the whole course grade.

Step-by-Step: Calculating the Current Grade

- 1

Average the Assignments
Add up all 4 assignment scores and divide by 4. This gives the `assignment_average`.
- 2

Drop the Lowest Quiz
The program finds the smallest quiz score (`min(quizzes)`) and removes it from the list — a common "drop your worst score" policy. Only the remaining 4 quizzes count.
- 3

Average the Remaining Quizzes, Then Convert
The remaining quiz scores (out of 10) are averaged, then multiplied by 10 to turn them into a percentage (`quiz_percentage`), so they're on the same 0-100 scale as everything else.
- 4

Combine Everything Using the Weights
Each score is multiplied by its weight, and the results are added together. Since the final exam hasn't happened yet, only assignments + quizzes + midterm weights (0.25 + 0.20 + 0.25 = 0.70) are used so far.
- 5

Scale It Back Up to 100%
Since only 70% of the grade has been earned so far, the program divides by 0.70 so the "current grade" is shown fairly, as if graded out of 100 — not out of 70.



WORKED EXAMPLE, WITH THESE NUMBERS	
CALCULATION	RESULT
Assignment average	$(88+94+91+85) / 4 = 89.50\%$
Lowest quiz dropped	7
Remaining quiz average	$(9+10+8+9) / 4 = 9.00 / 10$
Quiz percentage	$9.00 \times 10 = 90.00\%$
Midterm	82%
Current grade	$\approx 87.11\%$

Working Out the Required Final Exam Score

1 Ask for a Target Grade

The program asks the user, "What overall grade do you want?" — for example, 90%.

2 Work Backwards From the Target

It rearranges the grade formula to solve for the one unknown piece: the final exam score. In simple terms: *"How high does my final exam need to be to bring my overall average up to my target?"*

3 Check If It's Even Possible

If the required score is above 100%, the program tells the user the target is impossible — you can't score more than full marks.

Special Rule: Final Exam Can Replace the Midterm

THE LOGIC

If the required final exam score turns out to be **higher than the midterm score**, the program assumes a friendlier grading policy: the final exam grade replaces the midterm grade entirely (since the student did better on the final).

What changes: The midterm's 25% weight and the final's 30% weight are merged into one combined weight (55%), and the required final score is recalculated using this new, larger weight — making the required score easier to reach.

WHY THIS MATTERS

Without this rule, a student could need an unrealistically high final exam score. With the rule, the program recognizes that scoring extremely well on the final is impressive enough to make up for a weaker midterm — and adjusts the math accordingly.

What Gets Printed on Screen

CURRENT GRADE REPORT

- Assignment average
- Which quiz was dropped
- Quiz average (out of 10 and as %)
- Midterm score
- Final current grade

FINAL EXAM TARGET REPORT

- The score needed on the final exam
- Whether the "midterm replacement" rule applied
- A clear message if the target isn't reachable

Quick Glossary

TERM	SIMPLE MEANING
Weight	How much a category counts toward the final grade
Average	Adding up numbers and dividing by how many there are
Drop the lowest	Ignoring the worst score so it doesn't hurt the average
Completed weight	The portion of the grade already "locked in" before the final exam
Required final score	The exact percentage needed on the final to hit the target grade

Bottom line: This program takes the guesswork out of grade planning — it tells a student exactly where they stand and exactly what they need to do next.